DO NOW — RECALL

Lesson 102 · Choosing RISC, CISC, GPU and Multicore Sy

Name: _____ Class: _____ Date: _____

1. What does RISC primarily describe?

2. What does a GPU excel at?

3. Can a RISC CPU be multicore?

4. Graph edges A–B, A–C, B–D: BFS from A with B before C?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A reduced/regular instruction-set design approach.

2. Model answer

Large quantities of similar parallel operations.

3. Model answer

Yes. ISA and core count are independent characteristics.

4. Model answer

A,B,C,D.

By the end, you can…

• RISC/CISC and CPU/GPU are not a single four-way classification: a processor's ISA, core organisation and accelerators are different design properties.

• A graphics or machine-learning workload with abundant parallel operations may benefit from a GPU; sequential/control-intensive jobs suit general CPU cores.

• Choose based on power, energy budget, latency, software compatibility, transfer overhead and ability to divide the workload; support claims with mechanism.

Vocabulary — know these terms

ISA — Instruction Set Architecture.

accelerator — Hardware specialised for particular workloads.

latency — Delay of a task.

throughput — Amount of work per unit time.

NEW LEARNING

Read, annotate and explain

Understand

RISC/CISC and CPU/GPU are not a single four-way classification: a processor's ISA, core organisation and accelerators are different design properties.

Apply

A graphics or machine-learning workload with abundant parallel operations may benefit from a GPU; sequential/control-intensive jobs suit general CPU cores.

Evaluate

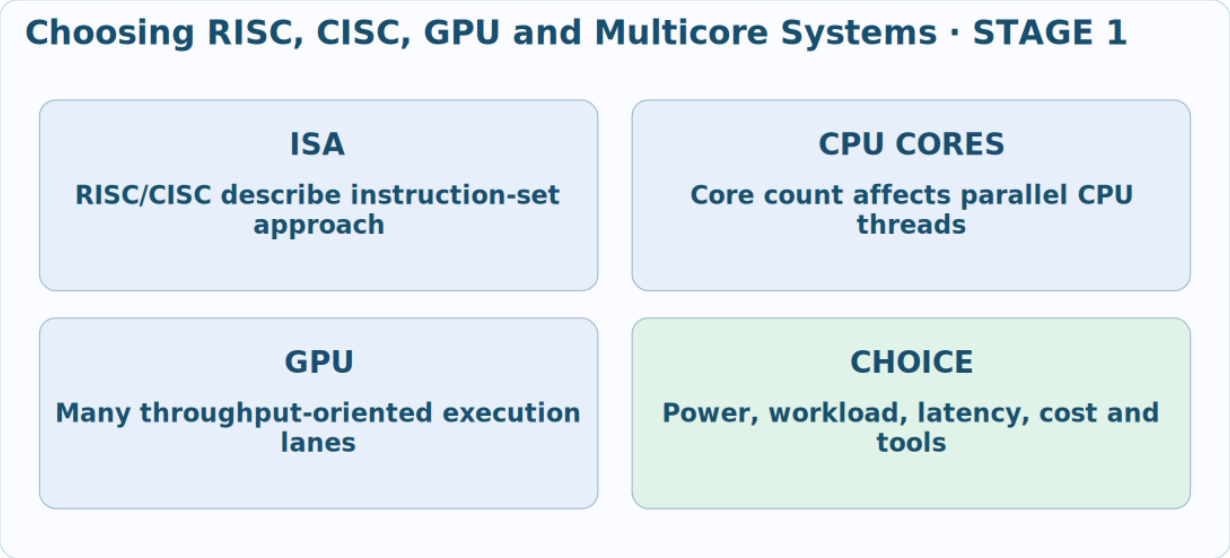
Choose based on power, energy budget, latency, software compatibility, transfer overhead and ability to divide the workload; support claims with mechanism.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

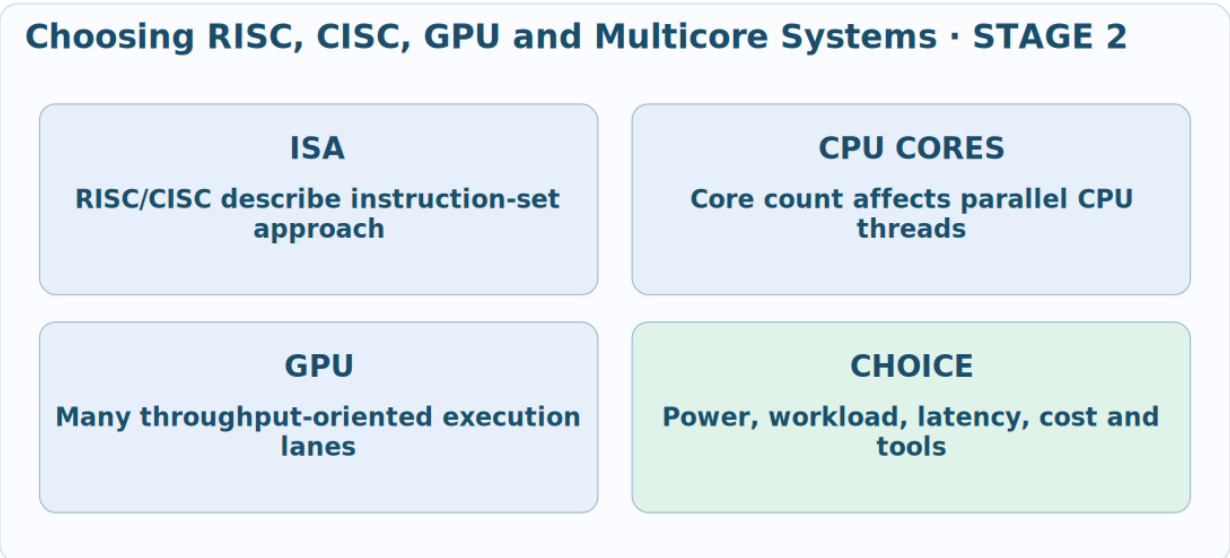
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Choosing RISC, CISC, GPU and Multicore Systems · STAGE 1; ISA: RISC/CISC describe instruction-set approach

New Learning

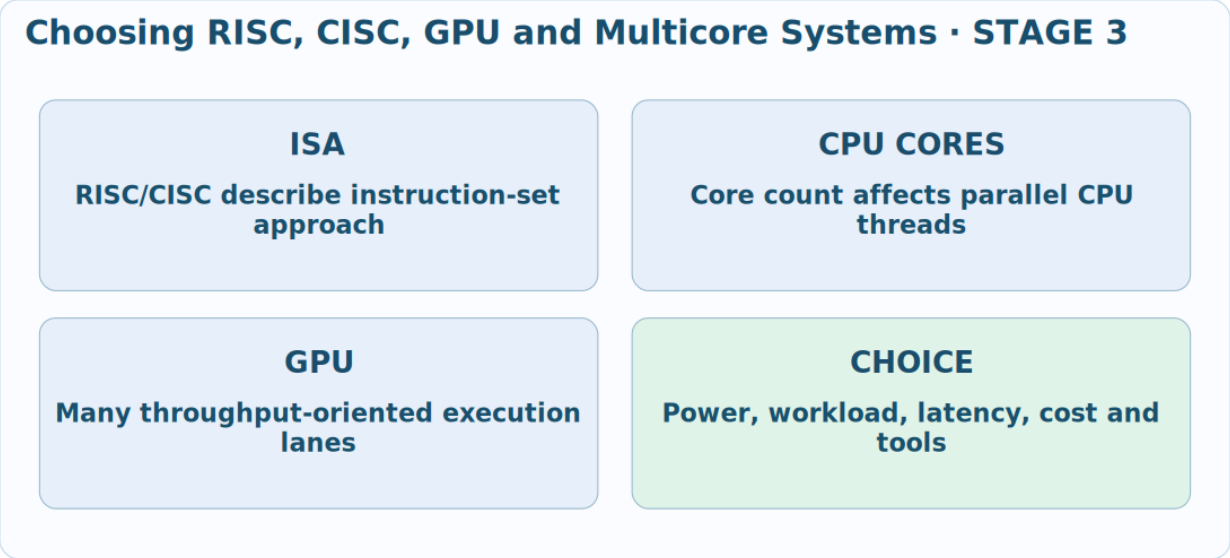


Choosing RISC, CISC, GPU and Multicore Systems · STAGE 2; ISA: RISC/CISC describe instruction-set approach; CPU CORES: Core count affects parallel CPU threads

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

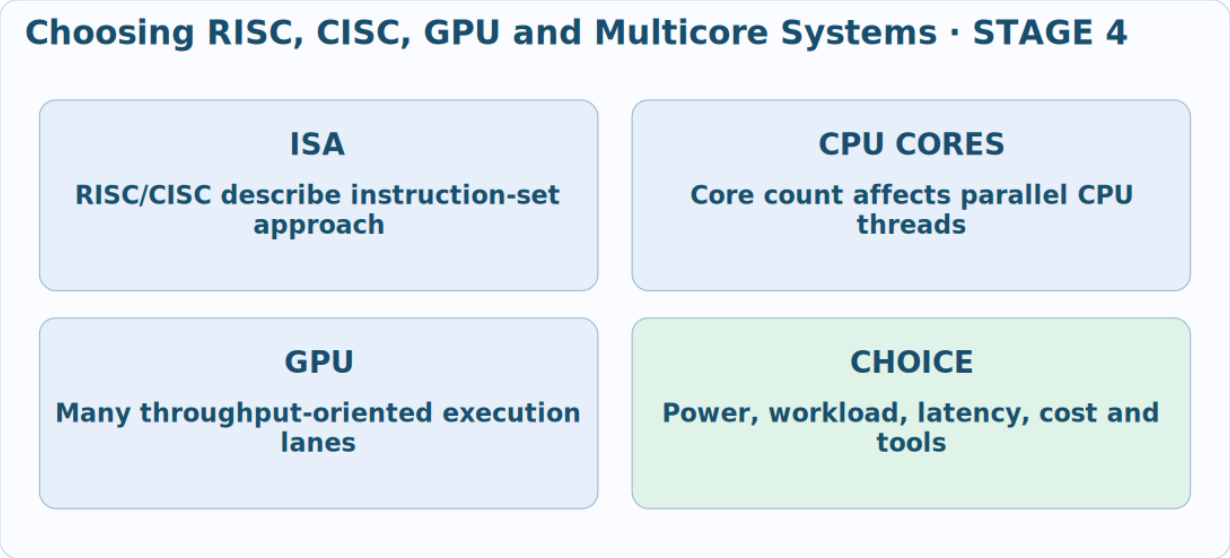


Choosing RISC, CISC, GPU and Multicore Systems · STAGE 3; ISA: RISC/CISC describe instruction-set approach; CPU CORES: Core count affects parallel CPU threads; GPU: Many throughput-oriented execution lanes

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Choosing RISC, CISC, GPU and Multicore Systems · STAGE 4; ISA: RISC/CISC describe instruction-set approach; CPU CORES: Core count affects parallel CPU threads; GPU: Many throughput-oriented execution lanes; CHOICE: Power, workload, latency, cost and tools

I DO — WORKED EXAMPLE

Follow the worked method

Recommend hardware for battery-powered sensor making a small control decision each second.

Explain the choice

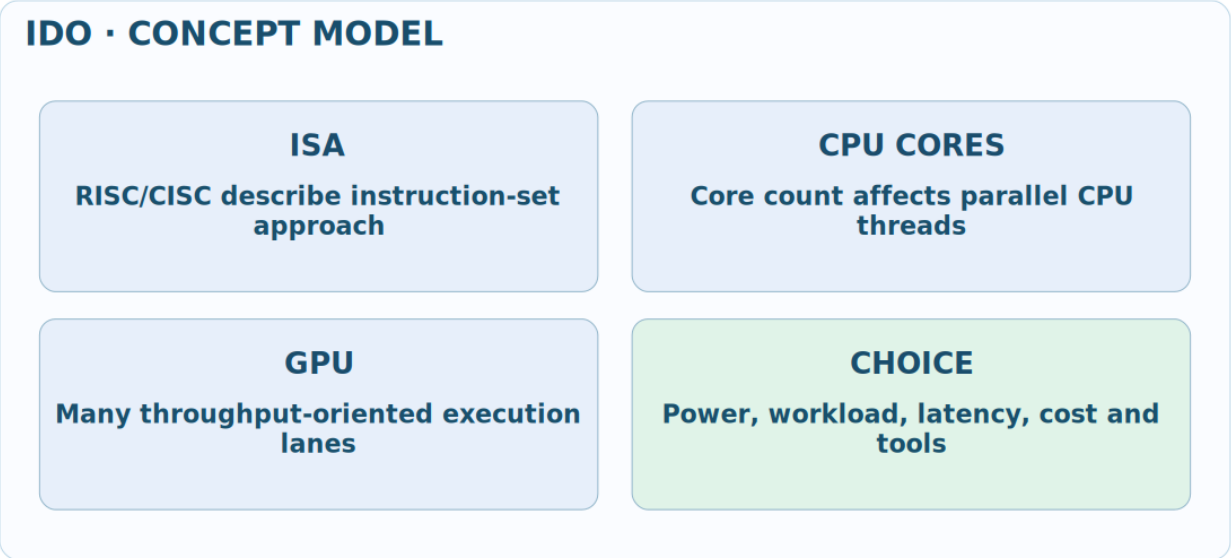
A low-power embedded CPU (often using a RISC architecture) is reasonable; no large GPU is needed for tiny sequential work.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Recommend hardware for battery-powered sensor making a small control decision each second.

A low-power embedded CPU (often using a RISC architecture) is reasonable; no large GPU is needed for tiny sequential work.



IDO · CONCEPT MODEL; Recommend hardware for battery-powered sensor making a small control decision each second.: A low-power embedded CPU (often using a RISC architecture) is reasonable; no large GPU is needed for tiny sequential work.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A studio renders millions of independent image elements per frame. Recommend computing hardware.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Explain why RISC/CISC and multicore are not competing mutually exclusive categories. [6]

Question 2

Select suitable processor features for (a) small battery sensor (b) large parallel graphics processing. Justify. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Evaluate CPU vs GPU investment for a school's interactive office tasks and high-volume video editing. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Every multicore processor must use a CISC instruction set.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

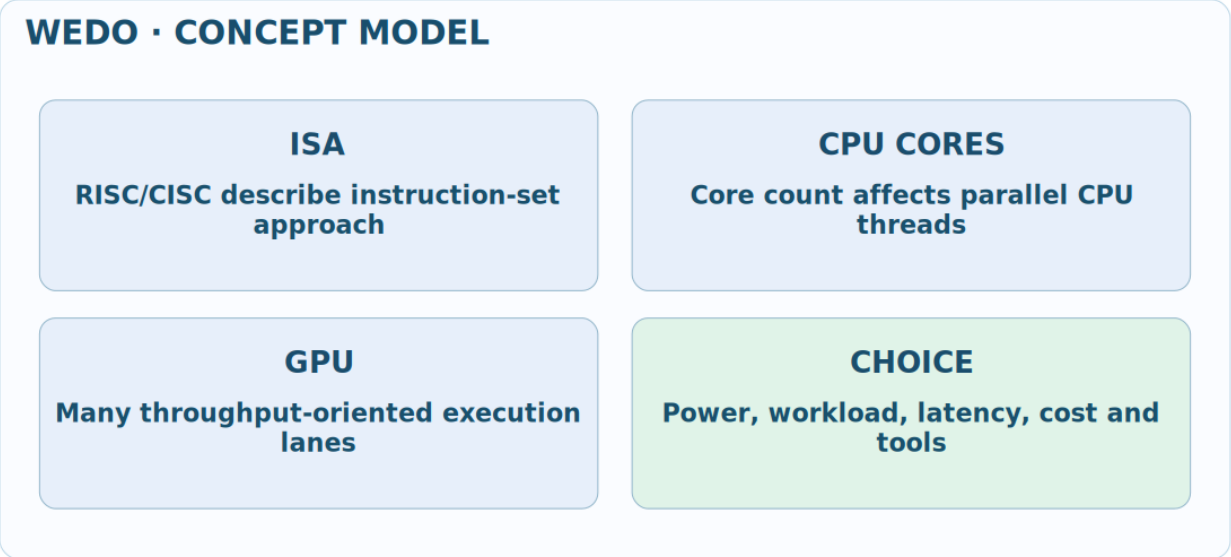
Explain the choice
A GPU can accelerate similar independent pixel/shading operations, paired with a CPU for control.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

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WEDO · CONCEPT MODEL; A studio renders millions of independent image elements per frame. Recommend computing hardware.: A GPU can accelerate similar independent pixel/shading operations, paired with a CPU for control.